

ANAKIN DEY

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EDUCATION

University of Illinois Urbana-Champaign

Bachelors of Science in Mathematics, Minor in Computer Science

May 2024

GPA: 3.93/4.00

EXPERIENCE

Undergraduate Research Assistant: Texas State University Math REU

Jun 2023 – Present

- Researching finite group classification using both mathematical and computational tools with [Prof. Yong Yang](#)
- Designing, proving, and implementing group theoretic algorithms using GAP, Magma, and SageMath
- Leveraging computer algebra systems on AWS EC2 to automate and manage the search for these structures which resulted in improving the testing and computing of these classifications from taking 2 days to just 3 hours

Software Engineering Intern: CME Group

Jun 2022 – Aug 2022

- Refactored large code bases in order to support multiple file formats and platforms for tracing and logging of messages, which resulted in the impact of tracing on system performance being reduced by over 20%
- Used Jenkins, XL Release and in-house tools to benchmark the performance impact of new formats
- Explored the use of dashboards and parsers to better process traces and give new insights on system performance

Undergraduate Research Assistant: [Learning, Decision, Control, Autonomy Lab](#)

Aug 2021 – Present

- Researching algorithms and heuristics for multi-target and multi-agent planning and movement under multiple constraints as well as deceptive and adversarial movement over graphs with [Prof. Melkior Ornik](#)
- Combining concepts from heuristic approximation algorithms, Markov decision processes, probability theory, geometry, and graph theory to design new and efficient heuristics and algorithms to apply to computational difficult problems
- Benchmarking algorithms in Python and using Matplotlib, NetworkX, and OSMnx to demonstrate informative statistics
- Maintaining near 100% code quality coverage with various tools such as Black, pytest, Flake8, Coverage.py, and mypy

Software Engineering Intern: SiteIQ

May 2021 – Aug 2021

- Improved the accuracy of status reports generated over 100 times per day from multiple fueling stations using Python and JavaScript by streaming and parsing data sources such as two-wire serial and ZMODEM files
- Identified multiple error codes and data that were not described in given documentation but present in 50% of devices and logged them for further analysis by the team

Course Assistant: UIUC

Jan 2021 – Present

- Course Assistant for Introduction to Computer Science (Jan 2021 - May 2021), Discrete Structures (Aug 2021 – Present), Introduction to Algorithms (Jan 2022 – Dec 2022) and Models of Computation (Jan 2023 - Aug 2023)

PUBLICATIONS

Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency

([IEEE ISC2 2022](#), [GitHub](#))

Anakin Dey, Melkior Ornik

Optimizing a Deterministic Measure of Counterdeceptiveness via Reattachment

(In Preparation)

Anakin Dey, Sam Ruggerio, Melkior Ornik

Classifying Primitive Solvable Permutation Groups of Rank 5 and 6

(In Preparation)

Anakin Dey, Kolton O'Neal, Duc Van Khanh Tran, Camron Upshur, Yong Yang

EXTRACURRICULARS

Founder & President: SIGma | Special Interest Group for Math and Algorithms | csttheory.org

- Founded club to teach theoretical computer science and mathematics to over 50 students of various backgrounds
- Creating weekly meetings covering ideas such as efficient [binary generation](#), [decidability](#) and Turing Machines, [state machines](#), and other [mathematical](#) concepts in a comprehensive and intuitive manner

Admin: SIGPwny | Special Interest Group for Security | sigpwny.com

- Hosted [multiple meetings](#) on cryptographical concepts including block ciphers, RSA, Diffie-Hellman, and elliptic-curve cryptography, complete with multiple challenges, to over 70 people per meeting
- Created cryptography challenges and reverse engineering for [UIUCTF 2021](#) / [2022](#) / [2023](#) for over 2000 participants and [Fall CTF 2021](#) / [2022](#) for over 300 participants

TECHNICAL SKILLS

Languages: Python + SageMath, GAP, Magma, Java, C++, JavaScript, TypeScript, Agda, C

Tools: Bash, Git, JUnit, mypy, pytest, Matplotlib, NetworkX, Numpy, Z3, Gurobi, GDB, Catch2, Pandas, Jira, Grafana